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# CLEANTECH: TRANSFORM WASTE MANAGEMENT WITH TRANSFER LEARNING

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## **HematoVision: Advanced Blood Cell Classification Using Transfer Learning**

### **1. Introduction**

HematoVision is a machine learning project developed during the SmartBridge Virtual Internship under the AI/ML domain. The goal is to classify blood cells into four types: Eosinophils, Lymphocytes, Monocytes, and Neutrophils using transfer learning techniques and deploy the model using a Flask web application.

### **2. Problem Statement**

Accurate and timely identification of blood cell types is crucial for diagnosing various diseases. Manual analysis is time-consuming and prone to human error. This project aims to automate the classification process using deep learning.

### **3. Objectives**

- To use transfer learning (MobileNetV2) to classify blood cell images.
- To visualize and evaluate the performance of the model.
- To integrate the trained model into a Flask web application.
- To deploy a simple UI for users to upload blood cell images and get predictions.

### **4. Tools and Technologies**

- Python
- TensorFlow / Keras
- OpenCV
- Flask
- Jupyter Notebook
- HTML/CSS (Milligram CSS)
- Git & GitHub

### **5. Dataset**

Source: Kaggle Blood Cell Dataset

Classes: Eosinophil, Lymphocyte, Monocyte, Neutrophil

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Total Images: 12,500 (augmented)

## 6. Project Workflow

### Step 1: Data Preprocessing

- Images resized to (224x224)
- Normalized using MobileNetV2's preprocess\_input

### Step 2: Model Building

- Used MobileNetV2 with frozen base layers
- Added custom dense layers

### Step 3: Model Evaluation

- Achieved ~89% accuracy
- Visualized training using loss and accuracy graphs
- Evaluated with classification report and confusion matrix

### Step 4: Saving the Model

- Saved the model as Blood Cell.h5

### Step 5: Web App using Flask

- Created home.html for image upload
- Created result.html to show prediction

## 7. Folder Structure

HematoVision/

app.py

Blood Cell.h5

static/

[uploaded images]

templates/

## HematoVision: Advanced Blood Cell Classification Using Transfer Learning

home.html

result.html

dataset/

[image folders]

report.pdf

demo\_video.mp4

### 8. Results

- Achieved ~89% classification accuracy
- Predicted all four blood cell types via web UI
- Users can upload images and view predictions instantly

### 9. Conclusion

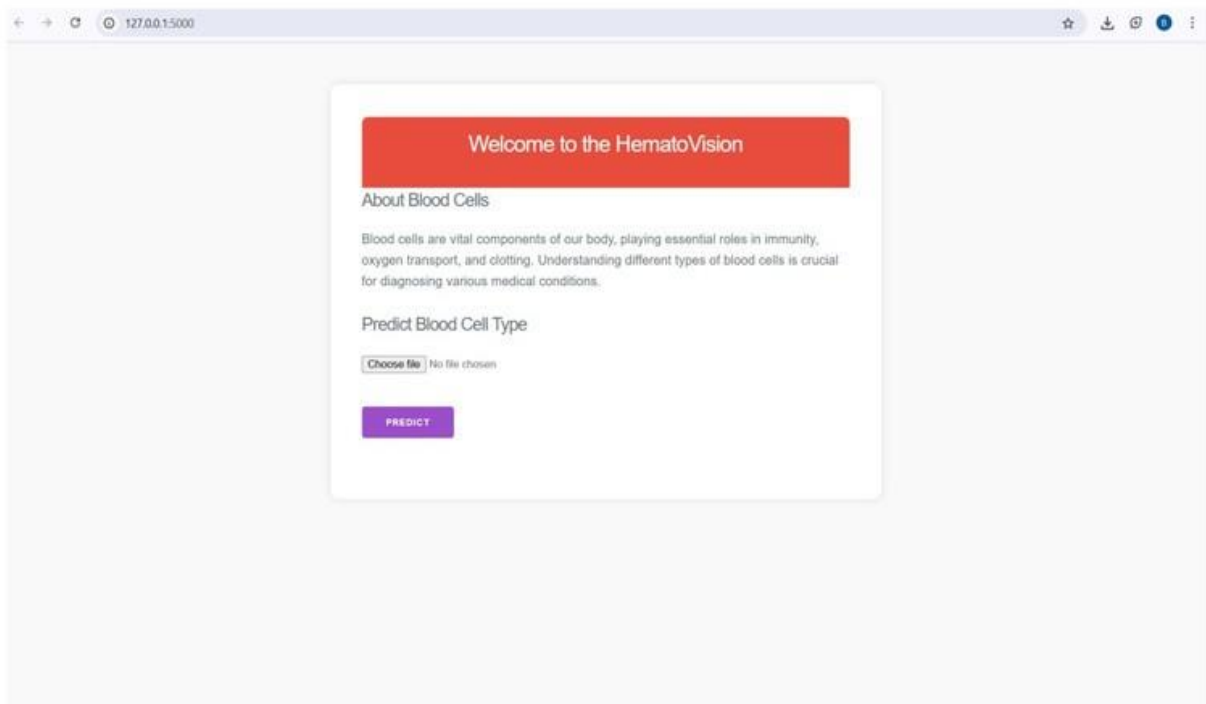
HematoVision demonstrates how transfer learning and Flask can be combined to build a real-time diagnostic tool. This helps reduce the burden on healthcare professionals by automating routine tasks.

### 10. Future Work

- Integrate additional cell types and datasets
- Add cloud deployment features
- Improve UI and add batch prediction support

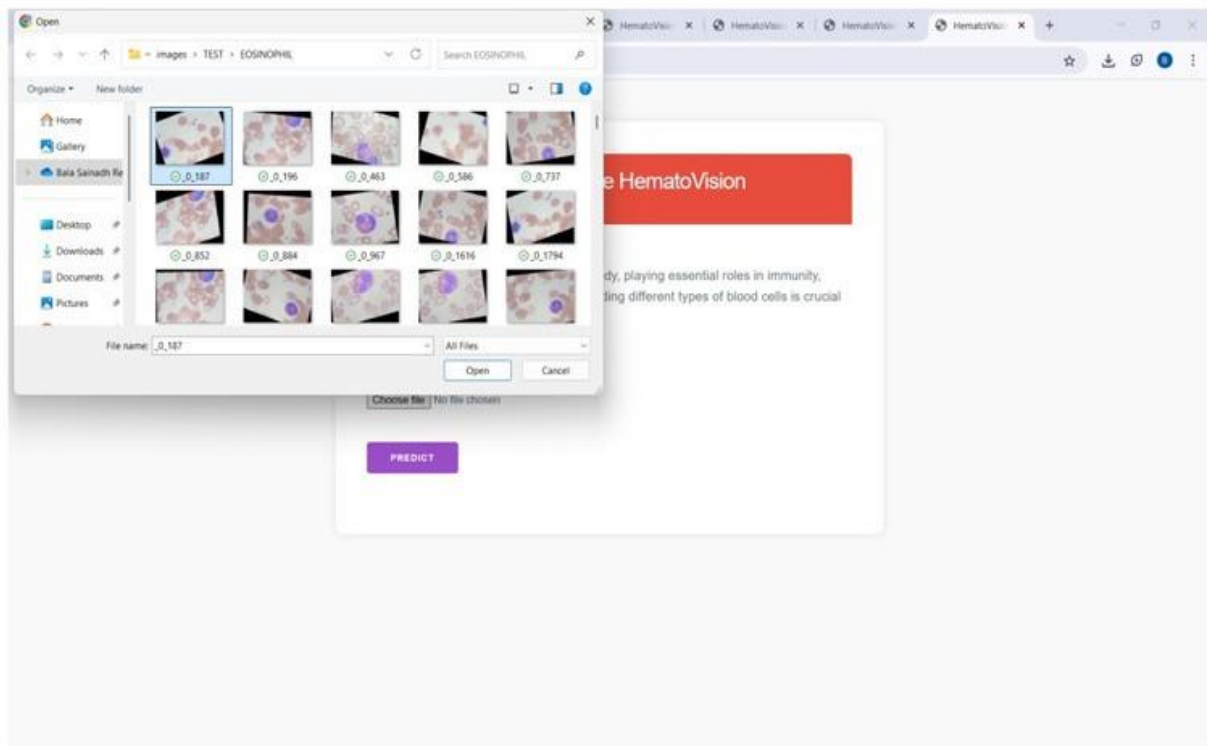
## Show Case ScreenShots:

### HematoVision: Advanced Blood Cell Classification Using Transfer Learning



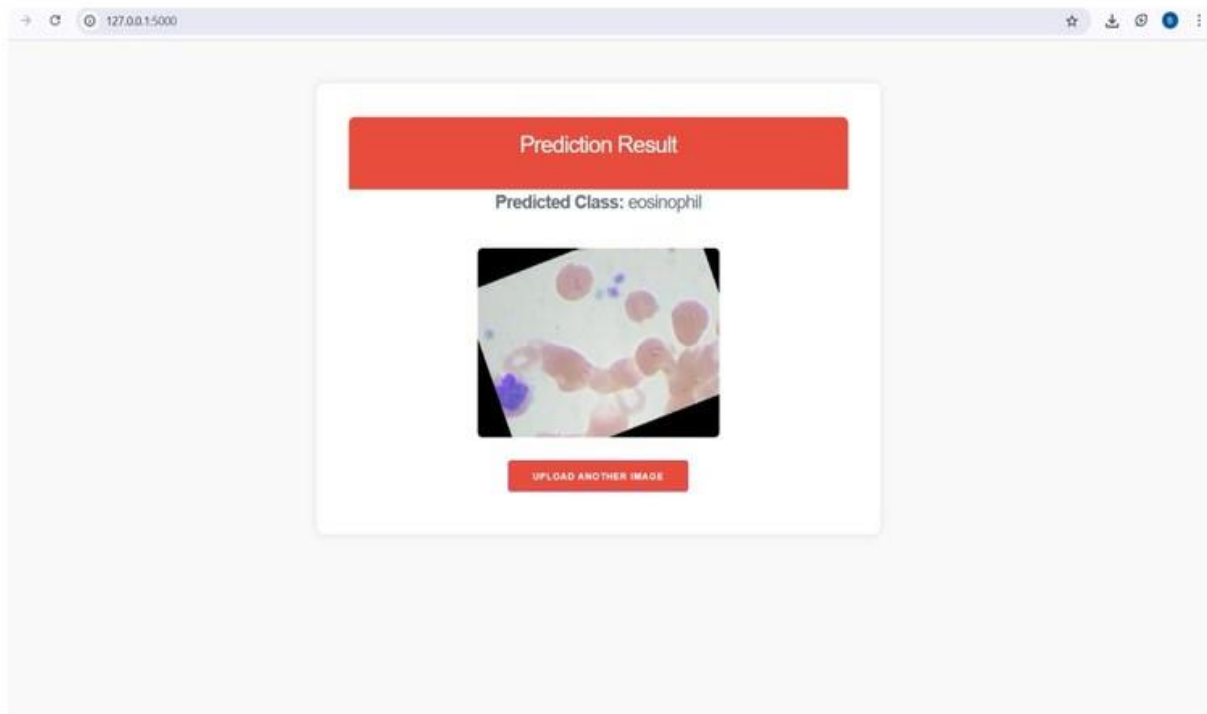
- Allows the user to upload microscopic blood cell images (JPEG/PNG).
- Ensures a user-friendly layout using Milligram CSS.
- Accepts files directly from local device storage.
- Prepares image for backend processing upon form submission.

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- ◆ Triggered when the user clicks the Predict button.
- ◆ Uploaded image is sent to Flask backend.
- ◆ The trained MobileNetV2 model processes and classifies the image.
- ◆ Prediction happens in real-time with fast response.

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- Displays the uploaded image preview alongside the predicted label.
- Predicted blood cell type: Eosinophil, Lymphocyte, Monocyte, or Neutrophil.
- Accurate result displayed using TensorFlow model's output.
- Provides visual confirmation to the user.